Computation of Set Tolerances with Applications to the Minimum Spanning Tree Problem

Dmitrii Panasenko, Umeå University, dmitrii.panasenko@umu.se
Gerold Jäger, Umeå University, gerold.jager@umu.se

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Overview

Definition and Examples

Define Combinatorial Sum Problem (CSP) $\mathcal{P} = (\mathcal{E}, D, c, f_c)$ as follows:

- $\mathcal{E}$: finite ground set of elements,
- D: set of feasible solutions,
- $c : \mathcal{E} \to \mathbb{R}$: cost function on $\mathcal{E}$,
- $f_c : D \to \mathbb{R}$: cost function on D with $f_c(S) = \sum_{e \in S} c(e)$ for $S \in D$.
- Let S^* be optimal, i.e., $f_c(S^*) = \min_{S \in D} f_c(S)$.
- Set $c^* := f_c(S^*)$.

In the following let $\mathcal{P} = (\mathcal{E}, D, c, f_c)$ be a CSP.

Famous examples of CSPs:

- Minimum Spanning Tree Problem (MSTP),
- Traveling Salesman Problem,
- Shortest Path Problem.

Illustrative Example

- $\mathcal{E} = \{a, b, d, e\}$,
- $D = \big\{\{a, d\}, \{b, d\}, \{a, b, d\}, \{a, e\}, \{e\}\big\}$,
- $c(a) = 1$, $c(b) = 1$, $c(d) = 5$, $c(e) = 8$,
- Optimal solutions: $\{a, d\}$ and $\{b, d\}$ with $c^* = 6$.

Notation

- $f_c(M) := \min\limits_{S \in M} \{f_c(S)\}$, where $M \subseteq D$.
- $D_-(E) := \{S \in D \mid E \cap S = \emptyset\}$.
- $D_+(E) := \{S \in D \mid E \subseteq S\}$.
- $D_\sim(E; E') := \{S \in D \mid E \cap S = \emptyset,\ E' \subseteq S\}$.

Let $e \in \mathcal{E}$.

Single upper tolerance $u'_{\mathcal{P}}(e)$:

Maximum by which $c(e)$ can be increased such that all optimal solutions S^* remain optimal.

Note that the costs of all other elements $\bar{e} \in \mathcal{E} \setminus \{e\}$ must remain unchanged.

Proposition 1 [Jäger, Turkensteen, 2026]

Let $e \in \mathcal{E}$. If e belongs to some optimal solution, then $u'_{\mathcal{P}}(e) = f_c(D_-(e)) - c^*$. Otherwise $u'_{\mathcal{P}}(e) = \infty$.

G. Jäger and M. Turkensteen, Extending the definition of single and set tolerances, *Operations Research Letters* **65**, 107407 (2026).

Illustrative Example Revisited

- $\mathcal{E} = \{a, b, d, e\}$,
- $D = \big\{\{a, d\}, \{b, d\}, \{a, b, d\}, \{a, e\}, \{e\}\big\}$,
- $c(a) = 1$, $c(b) = 1$, $c(d) = 5$, $c(e) = 8$,
- Optimal solutions: $\{a, d\}$ and $\{b, d\}$ with $c^* = 6$.

$\rightsquigarrow$ $u'_{\mathcal{P}}(a) = 0$, $u'_{\mathcal{P}}(b) = 0$, $u'_{\mathcal{P}}(d) = 2$, $u'_{\mathcal{P}}(e) = \infty$.

Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq \mathcal{E}$.
(Regular) set upper tolerance $u'_{\mathcal{P}}(E)$:
Maximum of all those α such that the costs of all elements $e \in E$ are not decreased,
the sum of all increases equals α,
and all optimal solutions S^* remain optimal.
Note that the costs of all other elements $\bar{e} \in \mathcal{E} \setminus E$ must remain unchanged.

Let $e \in \mathcal{E}$.
Single lower tolerance $l'_{\mathcal{P}}(e)$:
maximum by which the cost of e can be decreased such that the objective value of optimal solution remains the same.
Note that the costs of all other elements $\bar{e} \in \mathcal{E} \setminus \{e\}$ must remain unchanged.

Proposition 2 [Jäger, Turkensteen, 2026]

Let $e \in \mathcal{E}$. Then $l'_{\mathcal{P}}(e) = f_c(D_+(e)) - c^*$.

G. Jäger and M. Turkensteen, Extending the definition of single and set tolerances, *Operations Research Letters* **65**, 107407 (2026).

Illustrative Example Revisited

- $\mathcal{E} = \{a, b, d, e\}$,
- $D = \big\{\{a, d\}, \{b, d\}, \{a, b, d\}, \{a, e\}, \{e\}\big\}$,
- $c(a) = 1$, $c(b) = 1$, $c(d) = 5$, $c(e) = 8$,
- Optimal solutions: $\{a, d\}$ and $\{b, d\}$ with $c^* = 6$.

$\rightsquigarrow$ $l'_{\mathcal{P}}(a) = 0$, $l'_{\mathcal{P}}(b) = 0$, $l'_{\mathcal{P}}(d) = 0$, $l'_{\mathcal{P}}(e) = 2$.

Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq \mathcal{E}$.
(Regular) set lower tolerance $u'_{\mathcal{P}}(E)$:
maximum of all those α such that the costs of all elements $e \in E$ are not increased,
the sum of all decreases equals α,
and the optimal objective value remains the same.
Note that the costs of all elements $\bar{e} \in \mathcal{E} \setminus E$ must remain unchanged.

Set Upper Tolerances

In the following:

- Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq \mathcal{E}$.
- Let S^* be an optimal solution of $\mathcal{P}$.
- Let $E' = E \cap S^*$.
- Let $\alpha_1, \alpha_2, \ldots, \alpha_k \geq 0$ be the increases in the costs of the elements $e_1, e_2, \ldots, e_k$, respectively.

“Exact Upper Linear Program” (EUL)

$$\begin{aligned}
\text{maximize} \quad & \sum_{i=1}^{k} \alpha_i \\
\text{subject to} \quad & \\
\sum_{e_i \in E'} \alpha_i \quad &= \quad \sum_{e_i \in F} \alpha_i \quad \forall\, F \subseteq E \text{ with } f_c(D_\sim(E \setminus F; F)) = c^* \\
c^* + \sum_{e_i \in E'} \alpha_i \quad &\leq \quad f_c(D_\sim(E \setminus F; F)) + \sum_{e_i \in F} \alpha_i \\
& \qquad \forall\, F \subseteq E \text{ with } f_c(D_\sim(E \setminus F; F)) > c^* \\
\alpha_i \quad &\geq \quad 0 \quad \forall\, i \in [k]
\end{aligned}$$

Theorem 1

The optimal objective value of EUL is equal to the set upper tolerance $u'_{\mathcal{P}}(E)$.

Illustrative Example Revisited

- $\mathcal{E} = \{a, b, d, e\}$,
- $D = \big\{\{a, d\}, \{b, d\}, \{a, b, d\}, \{a, e\}, \{e\}\big\}$,
- $c(a) = 1$, $c(b) = 1$, $c(d) = 5$, $c(e) = 8$,
- Optimal solutions: $\{a, d\}$ and $\{b, d\}$ with $c^* = 6$.

Let $E = \{a, b\}$.
Choose $S^* = \{a, d\}$.
$\rightsquigarrow E' = E \cap S^* = \{a\}$.

Compute $D_\sim(E \setminus F; F)$ for each $F \subseteq E$ as follows:

- $F = \emptyset$: $D_\sim(\{a, b\}; \emptyset) = \big\{\{e\}\big\}$; $f_c(D_\sim(E \setminus F; F)) = 8$,
- $F = \{a\}$: $D_\sim(\{b\}; \{a\}) = \big\{\{a, d\}, \{a, e\}\big\}$; $f_c(D_\sim(E \setminus F; F)) = 6$,
- $F = \{b\}$: $D_\sim(\{a\}; \{b\}) = \big\{\{b, d\}\big\}$; $f_c(D_\sim(E \setminus F; F)) = 6$,
- $F = \{a, b\}$: $D_\sim(\emptyset; \{a, b\}) = \big\{\{a, b, d\}\big\}$; $f_c(D_\sim(E \setminus F; F)) = 7$.

Illustrative Example Revisited

EUL looks like:

$$
\begin{aligned}
\text{maximize} \quad & \alpha_1 + \alpha_2 \\
\text{subject to} \quad & \\
& 6 + \alpha_1 \leq 8 \\
& \alpha_1 = \alpha_1 \\
& \alpha_1 = \alpha_2 \\
& 6 + \alpha_1 \leq 7 + \alpha_1 + \alpha_2 \\
& \alpha_1 \geq 0 \\
& \alpha_2 \geq 0
\end{aligned}
$$

Illustrative Example Revisited

EUL looks like:

$$
\begin{aligned}
\text{maximize} \quad & \alpha_1 + \alpha_2 \\
\text{subject to} \quad & \\
& \alpha_1 \leq 2 \\
& \alpha_1 = \alpha_2 \\
& \alpha_2 \geq -1 \\
& \alpha_1 \geq 0 \\
& \alpha_2 \geq 0
\end{aligned}
$$

$\rightsquigarrow$ Optimal solution: $\alpha_1 = 2, \alpha_2 = 2$.

$\rightsquigarrow$ $u'_{\mathcal{P}}(E) = 2 + 2 = 4$.

Set Lower Tolerances

In the following:

- Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq \mathcal{E}$.
- Let $\alpha_1, \alpha_2, \ldots, \alpha_k$ be the decreases in the costs of $e_1, e_2, \ldots, e_k$, respectively.

Introduce three following LPs.

"Exact Lower Linear Program" (ELL)

$$
\begin{aligned}
\text{maximize} \quad & \sum_{i=1}^{k} \alpha_i \\
\text{subject to} \quad & \\
\sum_{e_i \in F} \alpha_i \quad & \leq \quad f_c(D_\sim(E \setminus F; F)) - c^* && \forall\, F \subseteq E \\
\alpha_i \quad & \geq \quad 0 && \forall\, i \in [k]
\end{aligned}
$$

"Include Lower Linear Program" (ILL)

$$
\begin{aligned}
\text{maximize} \quad & \sum_{i=1}^{k} \alpha_i \\
\text{subject to} \quad & \\
\sum_{e_i \in F} \alpha_i \quad & \leq \quad f_c(D_+(F)) - c^* & \forall\, F \subseteq E \\
\alpha_i \quad & \geq \quad 0 & \forall\, i \in [k]
\end{aligned}
$$

"Tolerance Lower Linear Program" (TLL)

$$
\begin{aligned}
\text{maximize} \quad & \sum_{i=1}^{k} \alpha_i \\
\text{subject to} \quad & \\
\sum_{i=1}^{k} \alpha_i \quad & \leq \quad f_c(D_+(E)) - c^* \\
\sum_{e_i \in F} \alpha_i \quad & \leq \quad l'_{\mathcal{P}}(F) && \forall\, F \subsetneq E,\ F \neq \emptyset \\
\alpha_i \quad & \geq \quad 0 && \forall\, i \in [k]
\end{aligned}
$$

Theorem 2

The optimal objective values of ELL, ILL and TLL are equal to the set lower tolerance $l'_{\mathcal{P}}(E)$.

Illustrative Example Revisited

- $\mathcal{E} = \{a, b, d, e\}$,
- $D = \big\{\{a, d\}, \{b, d\}, \{a, b, d\}, \{a, e\}, \{e\}\big\}$,
- $c(a) = 1$, $c(b) = 1$, $c(d) = 5$, $c(e) = 8$,
- Optimal solutions: $\{a, d\}$ and $\{b, d\}$ with $c^* = 6$.

Let $E = \{a, e\}$. Then

- $l'_{\mathcal{P}}(a) = 0$,
- $l'_{\mathcal{P}}(e) = 2$,
- $D_+(E) = D_+(\{a, e\}) = \big\{\{a, e\}\big\}$,
- $f_c(D_+(E)) = 9$.

Illustrative Example Revisited

TLL looks like:

$$\begin{aligned} \text{maximize} \quad & \alpha_1 + \alpha_2 \\ \text{subject to} \quad & \\ & \alpha_1 + \alpha_2 \leq 9 - 6 = 3 \\ & \alpha_1 \leq 0 \\ & \alpha_2 \leq 2 \\ & \alpha_1 \geq 0 \\ & \alpha_2 \geq 0 \end{aligned}$$

$\rightsquigarrow$ Optimal solution: $\alpha_1 = 0$, $\alpha_2 = 2$.
$\rightsquigarrow$ $l'_{\mathcal{P}}(E) = 0 + 2 = 2$.

Recursive Approach for Set Lower Tolerances

ILL and TLL allow construction of a recursive algorithm for computing the lower tolerances of all $E \subseteq \mathcal{E}$ as follows.

- Compute tolerances for sets of increasing cardinality, starting from $|E| = 1$.
- For each $E \subseteq \mathcal{E}$, evaluate $f_c(D_+(E)) - c^*$, and solve corresponding ILL or TLL.

$\rightsquigarrow$ This is sufficient since all values required for smaller subsets have already been computed in previous iterations.

Set Tolerances for Sets of Cardinality 2

Theorem 3

Let $E = \{e_1, e_2\} \subseteq \mathcal{E}$. Then

(a) If both e_1 and e_2 belong to the same optimal solution of $\mathcal{P}$, then
$$u'_{\mathcal{P}}(E) = \min\left\{u'_{\mathcal{P}}(e_1) + u'_{\mathcal{P}}(e_2), f_c(D_-(E)) - c^*\right\}.$$

(b) If e_1 and e_2 each belong to some optimal solution of $\mathcal{P}$, but not to the same optimal solution, then
$$u'_{\mathcal{P}}(E) = 2 \cdot \left(f_c(D_-(E)) - c^*\right).$$

(c) If e_1 or e_2 does not belong to any optimal solution of $\mathcal{P}$, then
$$u'_{\mathcal{P}}(E) = \infty.$$

Theorem 4

Let $E = \{e_1, e_2\} \subseteq \mathcal{E}$. Then
$$l'_{\mathcal{P}}(E) = \min\left\{l'_{\mathcal{P}}(e_1) + l'_{\mathcal{P}}(e_2),\ f_c(D_+(E)) - c^*\right\}.$$

Theorem 5

Let $E \subseteq \mathcal{E}$ and $E = \bigcup_{i=1}^{m} E_i$ an arbitrary partition. Then

$$l'_{\mathcal{P}}(E) \leq \sum_{i=1}^{m} l'_{\mathcal{P}}(E_i).$$

Theorem 6

Let $E = \{e_1, e_2, \ldots, e_k\} \subseteq \mathcal{E}$ and $s \in \{1, 2, \ldots, k-1\}$. Then

$$l'_{\mathcal{P}}(E) \leq \frac{1}{\binom{k-1}{s-1}} \sum_{F \subseteq E, |F| = s} l'_{\mathcal{P}}(F).$$

Definition

Minimum spanning tree problem (MSTP):
problem of finding a spanning tree of minimum weight in a given weighted connected undirected graph.

Formally:
Let Γ be a connected undirected graph.
MSTP is a CSP $\mathcal{P} = (\mathcal{E}, D, c, f_c)$, where

- $\mathcal{E} = E(\Gamma)$,
- D is the set of spanning trees in Γ,
- $c : E(\Gamma) \to \mathbb{R}$,
- $f_c(T) = \sum_{e \in E(T)} c(e) \; \forall \, T \in D$.

Single Tolerances

- Recursive method for computing single upper tolerances for the MSTP in $\mathcal{O}(n^2)$ time, where n is the number of vertices.
  $\rightsquigarrow$ [Chin, Houck, 1977]
- Method for computing single lower tolerances for the MSTP with the same complexity.
  $\rightsquigarrow$ [Helsgaun, 2000]
- To our knowledge, only single toleances have been considered for this problem, but not set tolerances.

F. Chin and D. Houck, Algorithms for updating minimal spanning trees, *Journal of Computer and System Sciences* **16**, 334–344 (1978).

K. Helsgaun, An effective implementation of the Lin-Kernighan traveling salesman heuristic, *European Journal of Operations Research* **126**(1), 106–130 (2000).

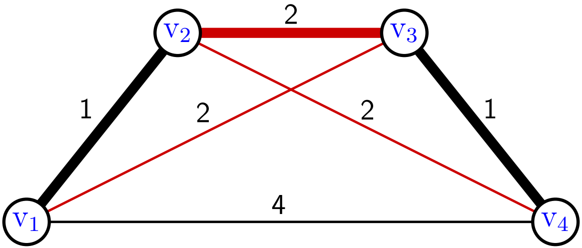
Theorem 8

Let $E = \{e_1, e_2, \dots, e_k\} \subseteq E(\Gamma)$. Then $u'_{\mathcal{P}}(E) \geq \sum_{i=1}^{k} u'_{\mathcal{P}}(e_i)$.

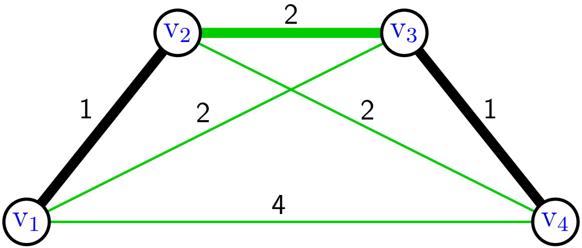
Theorem 9

Let $E = \{e_1, e_2, \dots, e_k\} \subseteq E(\Gamma)$. Then $l'_{\mathcal{P}}(E) = \sum_{i=1}^{k} l'_{\mathcal{P}}(e_i)$.

Example: Upper Tolerances



Example: Lower Tolerances



Future Work

Future Work

- Develop an algorithm for computing exact value of set upper tolerance for MSTP.
- Apply the extended theory of set tolerances to polynomially time solvable CSPs, like the Shortest Path Problem and the Linear Assignment Problem.
- Use the computation of set tolerances in heuristics to $\mathcal{NP}$-hard CSPs, like the Traveling Salesman Problem, as similar was done for single tolerances.

Thanks for your attention!